21/2/26
Saturday

CO3

Session - 16 & 17

Integral Calculus

Gamma function:-

$$\int_0^\infty t^{n-1} e^{-t}\, dt$$

Properties:-

① $\Gamma(n+1) = n\Gamma n$ } n is rational number

② $\Gamma n = \dfrac{\Gamma n+1}{n}$

③ $\Gamma(n+1) = n!$

④ $\Gamma\frac{1}{2} = \sqrt{\pi}$

⑤ $\Gamma(1-n) \cdot \Gamma n = \dfrac{\pi}{\sin n\pi}$

$$= \int_0^\infty t^{3} e^{-t}\, dt$$

$$= \Gamma 4$$

$$= \Gamma 3+1$$

$$= 3! = 6$$

$\Gamma\frac{9}{2} = \Gamma n+1 = n\Gamma n$

(or) $\Gamma n = \dfrac{\Gamma n+1}{n}$

$n+1 = \frac{9}{2}$

$n = \frac{9}{2} - 1$

$= 7/2$

$\Gamma\frac{9}{2} = \Gamma 7/2 + 1$

$= \frac{7}{2}\Gamma\frac{7}{2}$

$7/2 = \frac{5}{2} + 1$

$\Gamma\frac{7}{2} = \Gamma\frac{5}{2} + 1$

$= \frac{5}{2}\Gamma\frac{5}{2}$

$\Gamma 5/2 = \Gamma 3/2 + 1$

$= \frac{3}{2}\Gamma\frac{3}{2}$

$\Gamma\frac{3}{2} = \Gamma\frac{1}{2} + 1$

$= \frac{1}{2}\Gamma\frac{1}{2}$

$$\Gamma\frac{9}{2} = \frac{7}{2} \times \frac{5}{2} \times \frac{3}{2} \times \frac{1}{2}\sqrt{\pi}$$

(2) $\Gamma 6 = \Gamma 5+1 = 5!$ $(\Gamma n+1 = n!)$

(c) $\Gamma -\frac{7}{2}$

$$\Gamma -\frac{7}{2} = \frac{\Gamma -5/2}{-7/2} \qquad \Gamma -\frac{3}{2} = \frac{\Gamma -1/2}{-3/2}$$

$$\Gamma n+1 = n\Gamma n$$

$$\Gamma n = \frac{\Gamma n+1}{n} \qquad \Gamma -\frac{5}{2} = \frac{\Gamma -5/2+1}{-5/2} \qquad \Gamma -\frac{1}{2} = \frac{\Gamma -\frac{1}{2}+1}{-1/2}$$

$$n = -7/2 \qquad = \frac{\Gamma -3/2}{-5/2} \qquad = \frac{\Gamma 1/2}{-1/2}$$

$$\Gamma -7/2 = \frac{\Gamma -7/2+1}{-7/2} \qquad \Gamma -\frac{3}{2} = \frac{\Gamma -3/2+1}{-3/2} \qquad = \sqrt{\pi}/-1/2$$

$$\Gamma -\frac{7}{2} = \frac{\cancel{\Gamma\pi}\ \sqrt{\pi}}{-\frac{7}{2} \times -\frac{5}{2} \times \left(-\frac{3}{2}\right) \times \left(-\frac{1}{2}\right) \times \left(\frac{1}{2}\right)}$$

(3) Evaluate $\int_0^\infty x^5 e^{-x}\,dx$ using gamma form.

Sol:-

$$\Gamma n = \int_0^\infty x^{n-1} e^{-x}\,dx$$

$$n-1 = 5 \Rightarrow n = 6$$

$$\int_0^\infty x^{6-1} e^{-x}\,dx = \Gamma 6$$

$$= 5!$$

$$= 5 \times 4 \times 3 \times 2 \times 1$$

$$= 120$$

(4) Evaluate $\int_0^\infty e^{-2x} \cdot x^6\,dx$ using gamma fun^n.

Sol:-

Subtitute $2x = t$

$x = t/2$

$dx = 1/2\,dt$

$x-0 \Rightarrow t \to 0$

$x \to \infty \Rightarrow t - \infty$

$$I = \int_0^\infty e^{-t} \cdot \left(\frac{t}{2}\right)^6 dt$$

$$= \int_0^\infty e^{-t} t^6 \cdot \left(\frac{1}{2}\right)^6 dt$$

$$= \left(\frac{1}{2}\right)^7 \int_0^\infty e^{-t} t^6 dt$$

$$= \left(\frac{1}{2}\right)^7 \Gamma 7$$

$$= \left(\frac{1}{2}\right)^7 \times 6!$$

$$\beta(m,n) = \int_0^1 x^{m-1} (1-x)^{n-1} dx$$

$$= \int_1^0 (1-t)^{m-1} \cdot (t)^{n-1} (-dt)$$

$$= \int_0^1 (1-t)^{m-1} t^{n-1} dt = \beta(n,m)$$

Beta function:-

$$\beta(m,n) = \int_0^1 x^{(m-1)} \cdot (1-x)^{n-1} dx, \; m,n > 0$$

Beta function in trignometric form

$$\beta(m,n) = 2\int_0^{\pi/2} \sin^{(2m-1)}\theta \cdot \cos^{(2n-1)}\theta \, d\theta$$

$1-x = t \Rightarrow -dx = dt \Rightarrow x \to 0 \Rightarrow t \to 1$

$x = (1-t)$

$x \to 1 \Rightarrow t \to 0$

Relation between Beta and gamma funⁿ:-

$$\beta(m,n) = \frac{\Gamma m \, \Gamma n}{\Gamma(m+n)}$$

Properties:

① $\beta(m,n) = \beta(n,m)$

$$\int_a^b f(x) dx = -\int_b^a f(x) dx$$

(6) Compute $\beta(3,5)$, $\beta(\frac{1}{2},\frac{3}{2})$

Sol:-

$$\beta(3,5) = \frac{\Gamma 3\, \Gamma 5}{\Gamma 3+5} = \frac{2! \times 4!}{\Gamma 8} = \frac{2! \times 4!}{7!}$$

$$\beta\left(\frac{1}{2},\frac{3}{2}\right) = \frac{\Gamma\frac{1}{2}\cdot\Gamma\frac{3}{2}}{\Gamma\frac{1}{2}+\frac{3}{2}} = \frac{\Gamma\pi \cdot \frac{1}{2}\Gamma\pi}{\Gamma 2} = \frac{\pi}{2}$$

$$\beta(m,n) = \frac{\Gamma m\, \Gamma n}{(m+n)}$$

(7) use beta gamma fun^n to Evaluate

$$\int_0^1 x^4 (1-x)^3 dx$$

$$\beta(m,n) = \int_0^1 x^{(m-1)} (1-x)^{n-1} dx$$

$m-1 = 4 \Rightarrow m=5$

$n-1 = 3 \Rightarrow n=4$

$$\int_0^1 x^4 (1-x)^3 dx = \beta(5,4)$$

$$= \frac{\Gamma 5\, \Gamma 4}{\Gamma 5+4} = \frac{4! \times 3!}{8!}$$

Compute the integral $\int_0^{\pi/2} \sin^5 x \cdot \cos^3 dx$

Sol:-

$$\beta(m,n) = 2\int_0^{\pi/2} \sin\theta^{(2m-1)} \cdot \cos\theta^{(2n-1)} \cdot d\theta$$

compare $2m-1 = 5$ $\quad$ $2n-1=3$

$2m=6$ $\quad$ $2n-4$

$\boxed{m=3}$ $\quad$ $\boxed{n=2}$

$$I = \int_0^{\pi/2} \sin^5 x \cos^3 dx$$

$$I = \frac{1}{2} \beta(3,2)$$

$$= \frac{1}{2} \frac{\Gamma 3 \, \Gamma 2}{\Gamma 3+2} = \frac{1}{2} \frac{\Gamma 3 \, \Gamma 2}{\Gamma 5}$$

$$= \frac{1}{2} \frac{2! \times 1!}{4!}$$

Evaluate $\int_0^1 x^{1/2} (1-x)^{3/2} dx$

Sol:-

$$\beta\left(\frac{3}{2}, \frac{5}{2}\right) = \int_0^1 x^{3/2} (1-x)^{5/2} dx$$

$$= \frac{\Gamma \frac{3}{2} \, \Gamma \frac{5}{2}}{\Gamma \left(\frac{3}{2} + \frac{5}{2}\right)}$$

$$= \frac{\Gamma \frac{3}{2} \, \Gamma \frac{5}{2}}{\Gamma 8/2}$$

Tutorial-7

① compute the value $\Gamma \frac{13}{2}$ and $\Gamma \frac{-7}{2}$

Sol:- $\frac{13}{2} - 1 = n$

$$\Gamma \frac{11}{2} = n$$

$$\Gamma \left(\frac{11}{2} + 1\right) = \frac{11}{2} \Gamma \frac{11}{2}$$

$$\Rightarrow \Gamma \left(\frac{9}{2} + 1\right) = \frac{9}{2} \Gamma \frac{9}{2}$$

$$= \Gamma \left(\frac{7}{2} + 1\right) = \frac{7}{2} \Gamma \frac{7}{2}$$

$$= \Gamma \left(\frac{5}{2} + 1\right) = \frac{5}{2} \Gamma \frac{5}{2}$$

$$\Rightarrow \Gamma \left(\frac{3}{2} + 1\right) = \frac{3}{2} \Gamma \frac{3}{2}$$

$$\Rightarrow \Gamma \left(\frac{1}{2} + 1\right) = \frac{1}{2} \Gamma \frac{1}{2}$$

$$\Gamma \frac{13}{2} = \frac{11}{2} \times \frac{9}{2} \times \frac{7}{2} \times \frac{5}{2} \times \frac{3}{2} \times \frac{1}{2} \times \pi$$

$\Gamma \frac{-7}{2}$.

~~② Compute~~

$$n = \frac{\Gamma n+1}{n}$$

$$\Gamma \frac{-7}{2} = \frac{\Gamma \frac{-7}{2} + 1}{-7/2}$$

$$\Gamma \frac{-7}{2} = \frac{\Gamma \frac{-5}{2}}{-7/2}$$

$$\Gamma \frac{-5}{2} = \frac{\Gamma \frac{-5}{2} + 1}{-\frac{5}{2}}$$

$$\Gamma \frac{-3}{2} = \frac{\Gamma \frac{-3}{2} + 1}{-3/2}$$

$$\Gamma \frac{-3}{2} = \frac{\Gamma \frac{-1}{2}}{-3/2}$$

$$\Gamma \frac{-1}{2} = \frac{\Gamma \frac{-1}{2} - 1}{-1/2}$$

$\Gamma\frac{1}{2} = \sqrt{\pi}/-1/2$

$\therefore \Gamma\frac{-7}{2} = \dfrac{\sqrt{\pi}}{-7/2 \times -5/2 \times -3/2 \times -1/2 \times (1/2)}$

23/2/26 Monday

Section - 18

Evaluation of double Integral & it's Applications

$\int f(x)\,dx$ $\quad \iint f(x,y)\,dx\,dy$ → order of integration

(or)

$\iint f(x,y)\,dy\,dx$

Ex: $\int_0^2 \int_1^2 f(x,y)\,dx\,dy$, $\int_0^3 \int_{-1}^{1} (x^2+y^2)\,dy\,dx$

constant $\int x^2\,dx = \frac{x^4}{4}$

$\int_1^2 x^3\,dx = \left(\frac{x^4}{4}\right)_1^2 = \frac{1}{4}(2^4 - 1^4)$

Q) $\int_{x=0}^{1} \int_{y=0}^{3} 4xy^2\,dy\,dx$

Sol:- $\int_{x=0}^{1} 4x \int_0^3 y^2\,dy\,dx$

$= \int_{x=0}^{1} 4x \left[\frac{y^3}{3}\right]_0^3 dx$

$= \int_{x=0}^{1} 4x \frac{1}{3}\left[(3)^3 - 0^3\right] dx$

$= \int_{x=0}^{1} \frac{4}{3} \cdot x\,[27-0]\,dx$

$= \int_{x=0}^{1} \left[\frac{4}{\cancel{3}} \times \overset{9}{\cancel{27}} \times x\right] dx$

$I = \int_{36}^{1} x\,dx$

$= 36 \left[\frac{x^2}{2}\right]_0^1$

$= 36 \times \frac{1}{2}[1-0]$

$= 18$

Tutorial

2) $\int_0^3 \int_1^{4-y} (x+y)\, dx\, dy$

$\int c\,dx = cx$

$\int dy = cy$

$x^n dx = \frac{x^{n+1}}{n+1}$

$= \int_0^3 \left[\frac{x^2}{2} + yx\right]_1^{4-y} dy$

$= \int_0^3 \left[\frac{1}{2}(4-y)^2 + y(4-y) - \frac{1}{2}(1)^2 - y(1)\right] dy$

$= \int_0^3 \left[\frac{1}{2}\left[16 + y^2 - 8y\right] + 4y - y^2 - \frac{1}{2} - y\right] dy$

$= \int_0^3 \left[\frac{1}{2}\left[8 + \frac{1}{2}y^2 - 4y + 3y - y^2 - \frac{1}{2}\right]\right] dy$

$= \int_0^3 \left[-\frac{1}{2}y^2 - y + \frac{15}{2}\right] dy$

$= \left[-\frac{1}{2}\left(\frac{y^3}{3}\right) - \left(\frac{y^2}{2}\right) + \frac{15}{2}(y)\right]_0^3$

$= \left[-\frac{1}{6}(3)^3 - \frac{1}{2}(3)^2 + \frac{15}{2}(3) + \frac{1}{2}\left(\frac{0^3}{3}\right) + \left(\frac{0^2}{2}\right) + \frac{15}{2} \times 0\right]$

$= \left[-\frac{9}{2} - \frac{9}{2} + \frac{45}{2}\right]$

② Evaluate $\int_0^2 \int_0^{\sqrt{4-x^2}} (2x+y)\, dy\, dx$

Sol:

$I = \int_0^2 \int_0^{\sqrt{4-x^2}} (2x+y)\, dy\, dx$

$= \int_0^2 \left[2xy + \frac{y^2}{2}\right]_0^{\sqrt{4-x^2}} dx$

$= \int_0^2 \left[2x(\sqrt{4-x^2}) + \frac{1}{2}(\sqrt{4-x^2})^2 - 2x(0) - \frac{1}{2}(0)^2\right]$

$$I = \int_0^2 \Big[\underbrace{2x\sqrt{4-x^2}}_{I_1} + \underbrace{\tfrac{1}{2}(4-x^2)}_{I_2}\Big]dx$$

$$I_1 = \int_0^2 2x\sqrt{4-x^2}\,dx$$

Substitute $4-x^2 = t$

$-x^2 = (t-4)$

(or) $x = \sqrt{(4-t)}$

$\Rightarrow 4-x^2 = t$

diff w.r.to 'x'

$-2x\,dx = dt$

$4-x^2 = t$

Now limits:

$x \to 0 \Rightarrow t \to 4$

$x \to 2 \Rightarrow t \to 0$

$$I_1 = \int_4^0 \sqrt{t}\,(-dt)$$

$$\left(\int_a^b f(x)\,dx = -\int_b^a f(x)\,dx\right)$$

$$= \int_0^4 \sqrt{t}\,dt$$

$$= \left[\frac{t^{\frac{1}{2}+1}}{\frac{1}{2}+1}\right]_0^4$$

$$= \frac{2}{3}\left[t^{3/2}\right]_0^4$$

$$= \frac{2}{3}\left[4^{3/2} - 0\right]$$

$$= \frac{2}{3} \times (2^2)^{3/2} \Rightarrow \frac{16}{3}$$

$$I_2 = \frac{1}{2}\int_0^2 (4-x^2)\,dx$$

$$= \frac{1}{2}\left[4x - \frac{x^3}{3}\right]_0^2$$

$$= \frac{1}{2}\left[4\times 2 - \frac{1}{2}(2)^3 - 0(4) - \frac{1}{3}(0)^3\right]$$

$$= \frac{1}{2}\left[8 - \frac{8}{3}\right]$$

$$= \frac{1}{2}\left[\frac{24-8}{3}\right] = \frac{32}{3}$$

$$I = I_1 + I_2$$

$$= \frac{16}{3} + \frac{8}{3} = \frac{24}{3}$$

$$= 8$$

Basic Integration formula:-

① $\int x^n \, dx = \frac{x^{n+1}}{n+1}$

② $\int e^x \, dx = e^x$

③ $\int e^{ax} \, dx = \frac{e^{ax}}{a}$

④ $\int \sin x \, dx = -\cos x$

⑤ $\int \cos x \, dx = \sin x$

⑥ $\int \sec x \tan x \, dx = \sec x$

⑦ $\int \sec^2 x \, dx = \tan x$

⑧ $\int \frac{1}{1+x^2} \, dx = \tan^{-1} x$

⑨ $\int \frac{1}{\sqrt{1-x^2}} \, dx = \sin^{-1} x$

⑩ $\int \frac{1}{x} \, dx = \log x$

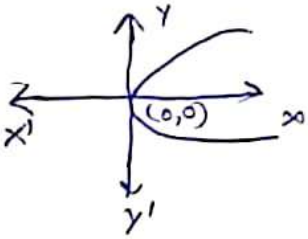
⑪ $\int \sin ax \, dx = -\frac{\cos ax}{a}$

⑫ $\int \cos ax \, dx = \frac{\sin ax}{a}$

⑬ $\int (ax+b)^n \, dx = \frac{1}{a} \frac{(ax+b)^{n+1}}{(n+1)}$

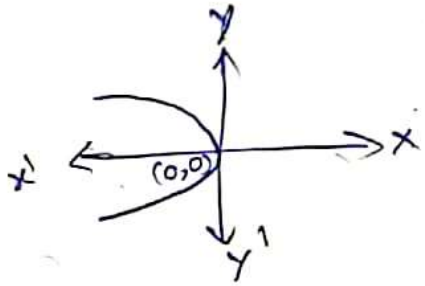
Some standard Curves:-

① Parabola (Right side)



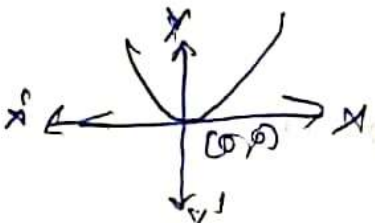
$y^2 = 4ax$

② Parabola (Left side)



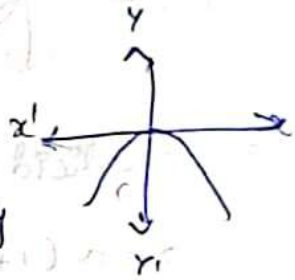
$y^2 = -4ax$

③ Parabola (upward)



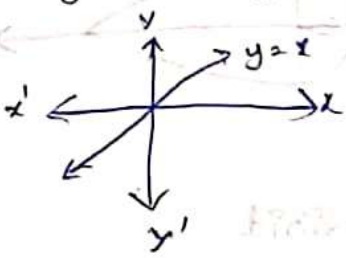
$x^2 = 4ay$

④ Parabola (downward)

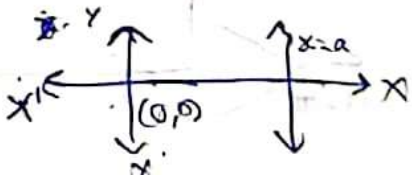


$x^2 = -4ay$

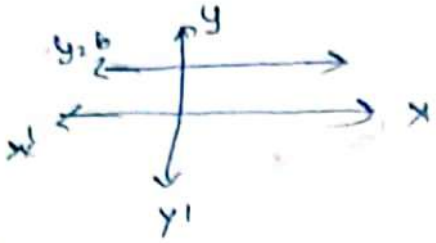
⑤ Straight line $y=x$



⑥ straigh line $x=a$ (parallel to y axis)

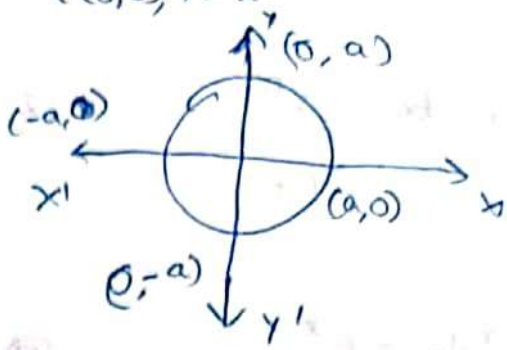


(7) Lin $y = b$



(8) Circle

$((0,0), r = a$

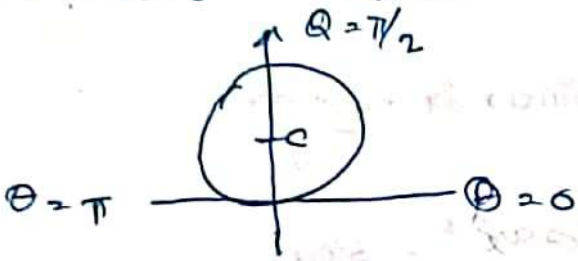


$x^2 + y^2 = a^2$

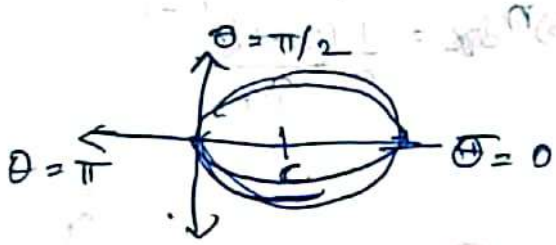
Polar curve (standard):-

Some standard polar curves.

(1) $r = a \sin\theta$ circle

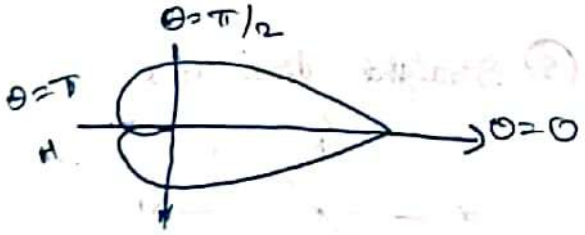


(2) $r = a \cos\theta$ circle



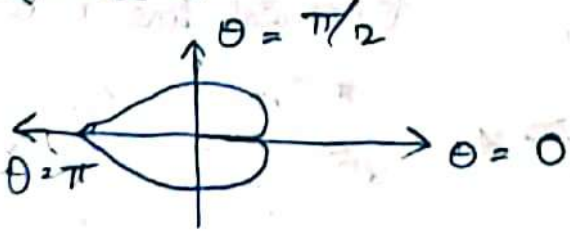
(3) Cardioid

$r = a(1 + \cos\theta)$



(4) Cardioid

$r = a(1 - \cos\theta)$

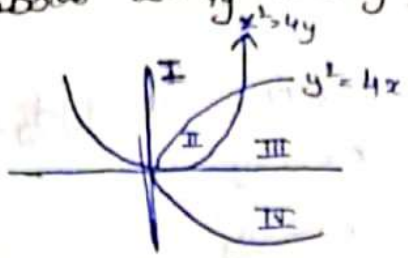


27/2/26
Friday

Session-18

4) Parabolas $x^2 = 4y$ and $y^2 = 4x$

Sol:



$\Rightarrow y = \frac{x^2}{4} \Rightarrow x^2 = 4y$

$y^2 = 4x$

$\left(\frac{x^2}{4}\right)^2 = 4x$

$x^3 = y^3$

$x = y$

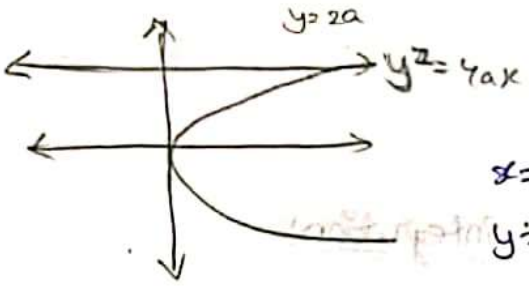
$\frac{x^4}{y^2} = 4x$

$y = x^2/4$

$y = 4$

Q) C is bounded by y-axis ordinate $y = 2a$, $y^2 = 4ax$

Sol:

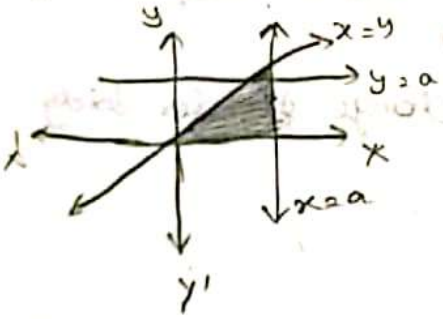


$x = 0$ $\quad x = y^2/4a$

$y = 0$ $\quad y = 2a$

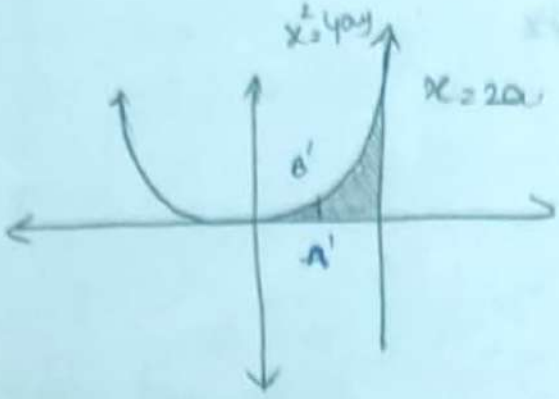
Q) Draw a region which is bounded by $x = y$, $x = a$, & $y = 0$.

Sol: $y = a$



Q) Determine the area of region which is bounded by x-axis, $x = 2a$ & the curve $x^2 = 4ay$

Soln



$\because x^2 = 4y$

$x = \sqrt{4y}$

$x = 2\sqrt{y}$

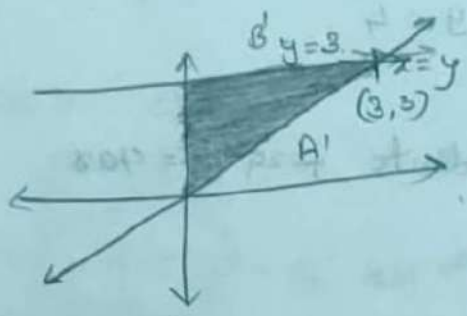
$\iint dy\, dx$

Q) change the order of Integration & Evalu

$$\int_0^3 \int_0^y (y^3 + x^2)\, dx\, dy$$

Soln limit dx : $x = 0$ to $x = y$

$y = 0$ to $y = 3$



$$\int_0^3 \int_x^3$$

28/2/26 Saturday

Session-19

Change of order of integration:-

$$\int_a^b \int_{g_1(x)}^{g_2(x)} f(x,y)\, dx\, dy \xrightarrow{\text{change of order}} \int_c^d \int_{h_1(y)}^{h_2(y)} f(x,y)\, dx\, dy \quad g_1(x) = x + e^x + 6x$$

order = dy dx

change of order dx dy

working process:-

1. Identify the given order
2. write the given limit and region
3. by change of order draw the ~~strip~~ parallel given limit variable in the region

(4) Identify the new limits and hence solve it

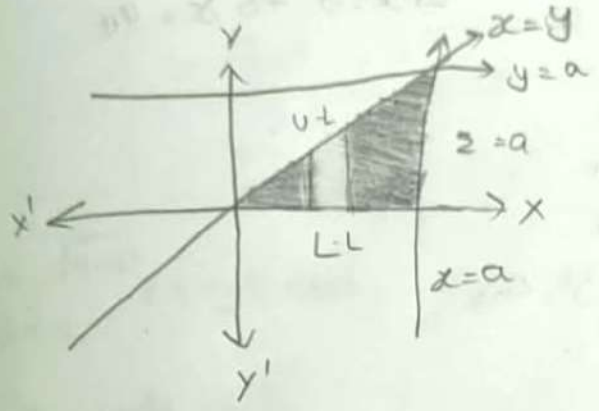
1) $\int_0^a \int_y^a \frac{4y}{x^2+y^2}\, dx\, dy$

Sol: given order is $dx\,dy$

by change of order $dy\,dx$

given limits : $y \le x \le a \Rightarrow x = y$ to $x = a$

$: 0 \le y \le a \Rightarrow y = 0$ to $y = a$



Y limits: L.L lies X-axis $Y = 0$

U.L lies on $x = y \Rightarrow y = x$

$0 \le y \le x$

x-limits: on x-axis.

$0 \le x \le a$

$$\int_0^a \int_y^a \frac{4x}{x^2+y^2}\, dx\, dy = \int_0^a \int_0^x \frac{4x}{x^2+y^2}\, dx\, dy$$

$$= \int_0^a 4x \int_0^x \frac{1}{x} \tan^{-1}\frac{y}{x}\Big]_0^x dx$$

$$\int \frac{1}{x^2+a^2}\, dx = \frac{1}{a}\tan^{-1}\frac{x}{a}$$

$\boxed{x = a}$

$$= \int_0^a 4x \left[\frac{1}{x}\tan^{-1}\frac{x}{x} - \frac{1}{0}\tan^{-1} 0\right] dx \qquad \tan^{-1} 1 = \frac{\pi}{4}$$

$$= \int_0^a 4x \frac{1}{x} \cdot \frac{\pi}{4}\, dx = \pi \cdot x\Big]_0^a = \pi(a-0) = \pi a$$

2) $\int_0^{4a}\int_{x^2/4a}^{2\sqrt{ax}} dy\,dx$

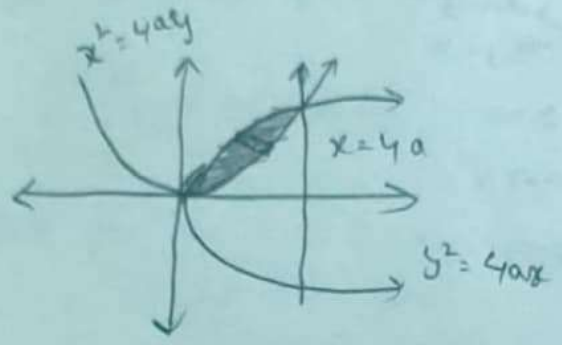
Sol: given order $dy\,dx$

change of order $dx\,dy$

given limits $\frac{x^2}{4a} \le y \le 2\sqrt{ax} \Rightarrow y = \frac{x^2}{4a}$ to $y = 2\sqrt{ax}$

$0 \le x \le 4a$ $\quad x^2 = 4ay$ to $y^2 = 4ax$

$\Rightarrow x = 0$ to $x = 4a$



x-limits:

L.L lies on $y^2 = 4ax \Rightarrow x = \frac{y^2}{4a}$

U.L lies on $x^2 = 4ay \Rightarrow x = \sqrt{4ay}$

y-limits:

on Y-axis $0 \le y \le 4a$

$$\int_0^{4a}\int_{\frac{x^2}{4a}}^{2\sqrt{ax}} dy\,dx = \int_0^{4a}\int_{\frac{y^2}{4a}}^{\sqrt{4ay}} dx\,dy$$

$$= \int_0^{4a} x\Big]_{y^2/4a}^{\sqrt{4ay}} dy$$

$$= \int_0^{4a} \left(\sqrt{4ay} - y^2/4a\right) dy$$

$$= \sqrt{4a}\cdot y^{3/2}/\tfrac{3}{2} - y^3/3\cdot 4a \Big]_0^{4a}$$

$\int\sqrt{x} = \frac{x^{3/2}}{3/2}$

$\int x^n = \frac{x^{n+1}}{n+1}$

$$= \sqrt{4a}\,\frac{(4a)^{3/2}}{3/2} - \frac{(4a)^3}{12a} - 0$$

$$= 2a^{1/2}\, 2^3 \cdot a^{3/2} \times \frac{2}{3} - \frac{\cancel{64a^3}^{\,16}}{\cancel{12}_{\,3}\,a}$$

$$\frac{32}{3}a^2 - \frac{16a^2}{3}$$

$$= \frac{16a^2}{3}$$

3. $\int_0^2 \int_0^{\sqrt{4-x^2}} (x+y)\,dx\,dy$

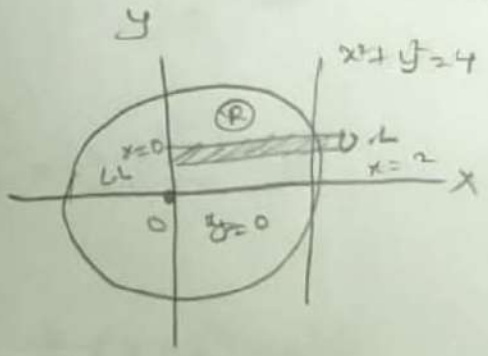
Sol:- given order $dx\,dy$.

change of order $dy\,dx$

given limits $0 \le y \le \sqrt{4-x^2} \Rightarrow y = 0$ to $y = \sqrt{4-x^2}$

$y = 0$ to $x^2 + y^2 = 4$

$\Rightarrow x = 0$ to $x = 2$



x limits:-

L.L lies on Y-axis $\Rightarrow x = 0$

U.L lies on $x^2 + y^2 = 4 \Rightarrow x = \sqrt{4-y^2}$

y-limits on Y-axis:

$0 \le y \le 2$

$$\therefore \int_0^2 \int_0^{\sqrt{4-x^2}} (x+y)\,dx\,dy = \int_0^2 \int^{\sqrt{4-y^2}} (x+y)\,dx\,dy.$$

$$= \int_0^2 \left[\frac{x^2}{2} + yx\right]_0^{\sqrt{4-y^2}} dy$$

$$= \int_0^2 \left[\frac{(\sqrt{4-y^2})^2}{2} + y\sqrt{4-y^2} - 0\right] dy$$

$$= \int_0^2 \left(\frac{4-y^2}{2} + y\sqrt{4-y^2}\right) dy$$

$\int f^n \cdot f' = \frac{f^{n+1}}{n+1}$

$$= \frac{4y - y^3/3}{2}\Big]_0^2 + \int_0^2 (4-y^2)^{1/2} \left(-\frac{2y}{-2}\right) dy$$

$$= \frac{8 - 8/3}{2} - 0 + \frac{(4-y^2)^{3/2}}{\frac{3}{2}(-2)}\Big]_0^2$$

$$\frac{16}{3\times 2} + \frac{0 - 4^{3/2}}{-3}$$

$$8/3 + 8/3$$

$$= \frac{16}{3}$$

4) $\int_0^2 \int_0^{2-y} xy \, dx \, dy$

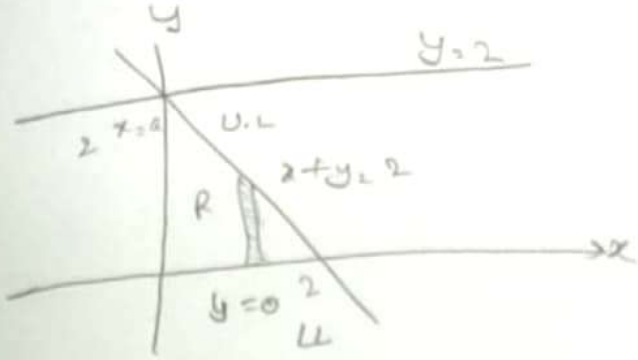
Sol:- $dx \, dy$

$dy \, dx$

$0 \le x \le 2-y \Rightarrow x=0$ to $x=2-y$

$0 \le y \le 2$ $\quad x=0$ to $x+y=2$

$\Rightarrow y=0$ to $y=2$



y-axis :- L.L lies on x-axis.

x-axis :

U.L lies ~~at~~ $y = 2 \Rightarrow y = 2 - x$

$$\boxed{0 \le y \le 2 - x}$$

$$\int_0^2 \int_0^{2-y} xy \, dx \, dy$$

$$= \int_0^2 \int_0^{2-x} xy \, dy \, dx$$

$$= \int_0^2 x \cdot \frac{y^2}{2} \Big]_0^{2-x} dx$$

$$= \int_0^2 x \frac{(2-x)^2 - 0}{2} dx$$

$$= \int_0^2 \frac{x}{2} (4 + x^2 - 4x) dx$$

$$= \int_0^2 (2x + \frac{x^3}{2} - 2x^2$$

$$= \cancel{2} \cdot \frac{x^2}{\cancel{2}} + \frac{x^4}{8} - \frac{2x^3}{3} \Big]_0^2$$

$$= 2^2 + \frac{2^4}{8} - \frac{2}{3} 2^3 - 0$$

$$= 4 + 2 - \frac{16}{3}$$

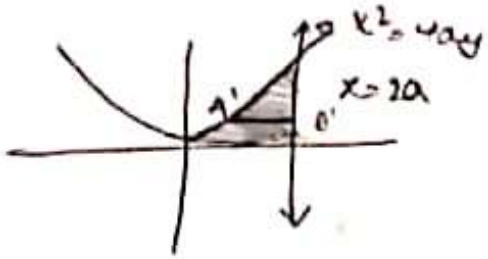
$$= 6 - \frac{16}{3} = \frac{18 - 16}{3} = \frac{2}{3}$$

2/3/22 Monday

Tutorial (8).

(6) Determine the area of the region bound by x-axis, $x = 2a$ and the curve $x^2 = 4ay$

Sol:



x-axis
$x = 2a$
$x^2 = 4ay$

$x^2 = 4ay$
$(2a)^2 = 4ay$
$4a^2 = 4ay$
$y = a$

Area $= \iint_1 dA$ $dA = dx\,dy$ (or) $dy\,dx$

$= \iint dx\,dy$

Lower limit of x: $x = 2\sqrt{ay}$
upp lim of x: $x = 2a$

lower limit y: $y = 0$
upp limit y: $y = a$

~~dA~~ $A = \int_0^a \int_{x=2\sqrt{ay}}^{2a} dx\,dy$

$= \int_0^a [x]_{2\sqrt{ay}}^{2a} dy$

$= \int_0^a [2a - 2\sqrt{a}\sqrt{y}]\,dy$

$= \left[2ay - 2\sqrt{a}\,\frac{y^{3/2}}{3/2}\right]_0^a$

(5) $y^2 = x$ and $x^2 = y$ Sketch the region R in the XY-plane bounded by the curves & determine its area.

Sol:

$\int_0^1 dx\, dy$ Lower limit $= y^2$, upper limit $= \sqrt{y}$

for $\iint_0 dy\,dx$ Lower limit $= x^2$, upper limit $= \sqrt{x}$

$x^2 = y, \; y^2 = x$

$(x^2)^2 = x$

$\Rightarrow x^4 = x$

$\Rightarrow x^4 - x = 0$

$x(x^3 - 1) = 0$

$x = 0, \; x = 1$

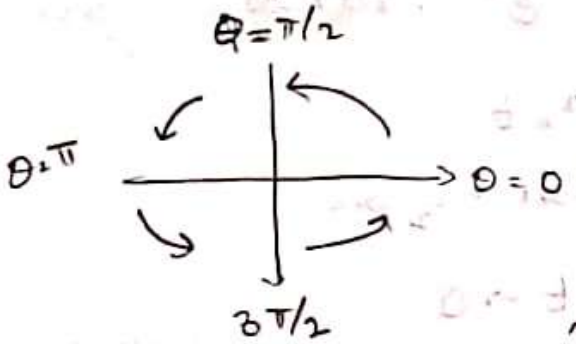
Session - 26:-

Evaluation of Double Integral by using change of variable

Polar coordinate system

$x = r\cos\theta, \; y = r\sin\theta$

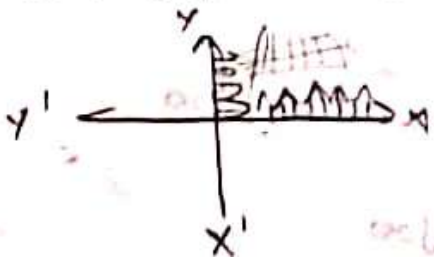
$dx\,dy$ (or) $dy\,dx = r.dr\,d\theta$



① $\int_0^\infty \int_0^\infty e^{-(x^2+y^2)}\,dx\,dy.$

Soln: x limits : $x=0$ to $x \to \infty$

y limits : $y=0$ to $y \to \infty$



The Ist quadrant is region of integration

$x = r\cos\theta$; $y = r\sin\theta$

$dx\,dy = r.dr.d\theta$

(x^2+y^2)

$f(x,y) = e^{-(x^2+y^2)}$

$= e^{-r^2(\cos)^2\theta + r^2(\sin)^2\theta}$

$= e^{-r^2}$

$f(x,y) = e^{-r^2}$

$\theta = 0$ to $\theta = \pi/2$

$r = 0$ to $r \to \infty$

$$Q = \int_0^{\infty}\int_0^{\infty} e^{-(x^2+y^2)}\,dx\,dy.$$

$$= \int_0^{\pi/2}\int_0^{\infty} e^{-r^2}.\,r\,dr\,d\theta$$

Substitute $r^2 = t$

$r\,dr = \frac{1}{2}\,dt$

$r \to 0 \Rightarrow t \to 0$

$r \to \infty \Rightarrow t \to \infty$

$$\int_0^{\pi/2}\int_0^{\infty} e^{-t}\left(\frac{1}{2}\,dt\right)d\theta$$

$$= \frac{1}{2}\int_0^{\pi/2}\left[-e^{-t}\right]_0^{\infty} d\theta$$

$$= \frac{1}{2}\int_0^{\pi/2}\left[-e^{-\infty} + e^{0}\right] d\theta$$

$e^{-\infty} = \frac{1}{e^{\infty}} = \frac{1}{\infty} = 0$

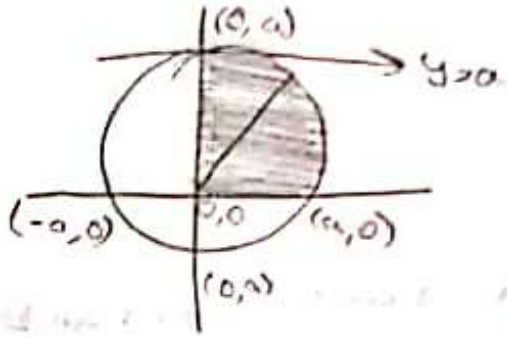
$$= \frac{1}{2}\int_0^{\pi/2}[0+1]\,d\theta$$

$$= \frac{1}{2}[\theta]_0^{\pi/2} = \frac{1}{2}\left(\frac{\pi}{2} - 0\right)$$

$$= \frac{\pi}{4}$$

2) $\int_0^a \int_0^{\sqrt{a^2-y^2}} (x^2+y^2)\, dx\, dy$

Sol: given limits.



x: $x=0$ to $x=\sqrt{a^2-y^2}$

$= x^2 = a^2 - y^2$

$x^2+y^2=a^2$

y: $y=0$ to $y=a$

$x = r\cos\theta$: $y = r\sin\theta$

$dx\,dy = r.dr\,d\theta$

$f(x,y) = (x^2+y^2)$

$= r^2\cos^2\theta + r^2\sin^2\theta$

$= r^2(\cos\theta + \sin^2\theta)$

$f(x,y) = r^2(1)$

limits of r: $r=0$ to $r=a$

limits of $\theta = 0$ to $\theta = \frac{\pi}{2}$

$$I = \int_0^a \int_0^{\sqrt{a^2-y^2}} (x^2+y^2)\, dx\, dy.$$

$$= \int_0^{\pi/2} \int_0^a r^2 . r\, dr\, d\theta$$

$$= \left(\int_0^{\pi/2} d\theta\right) \times \left(\int_0^a r^3\, dr\right)$$

$$= (\theta)_0^{\pi/2} \times \left(\frac{r^4}{4}\right)_0^a$$

$$= \frac{\pi}{2}\left(\frac{a^4}{4}\right)$$

$$= \frac{\pi a^4}{8}$$

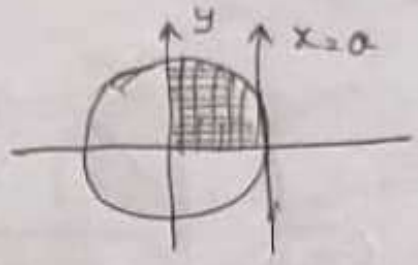
Q) $\int_0^a \int_0^{\sqrt{a^2-x^2}} y\sqrt{x^2+y^2}\, dx\, dy$

Sol:-

y limits : $y=0$ to $y=\sqrt{a^2-x^2}$

$y^2 = a^2 - x^2$ or $x^2+y^2=a^2$

x limits : $x=0$ to $x=a$



$x = r\cos\theta$; $y = r\sin\theta$

$dy\,dx = r\,dr\,d\theta$

$f(x,y) = y\sqrt{x^2+y^2}$

$= r\sin\theta\left(\sqrt{r^2\cos^2\theta + r^2\sin^2\theta}\right)$

$= r\sin\theta\,(r)$

$f(x,y) = r^2\sin\theta$

limits of r : $r=0$ to $r=a$

limits of θ : $\theta=0$ to $\theta=\pi/2$

$I = \int_0^{\pi/2}\int_0^a r^2\sin\theta\, r\,dr\,d\theta$

$= \left(\int_0^{\pi/2}\sin\theta\, d\theta\right) \times \left(\int_0^a r^3\, dr\right)$

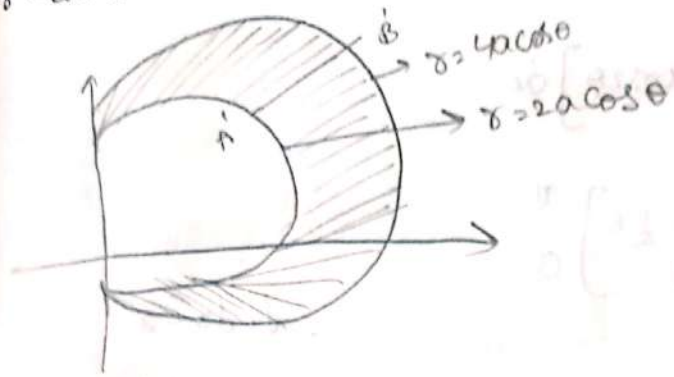
$= \frac{a^4}{4}$

21-12-16
Monday

Session-18

(5) $r = 2a\cos\theta$ & $r = 4a\cos\theta$

Sol:-



Area $= \iint dA$

$dA = dx\,dy$ (or) $dy\,dx$

In polar coordinates $dx\,dy = r\,dr\,d\theta$

Area $= \iint r\,dr\,d\theta$

θ limits: $\theta = 0$ to $\theta = \pi$

r limits: $r = 2a\cos\theta$ to $r = 4a\cos\theta$

$$A = \int_{\theta=0}^{\pi} \int_{r=2a\cos\theta}^{r=4a\cos\theta} r\,dr\,d\theta$$

$$= \int_0^{\pi} \left[\frac{r^2}{2}\right]_{2a\cos\theta}^{4a\cos\theta} d\theta$$

$$= \frac{1}{2}\int_0^{\pi} \left[16a^2\cos^2\theta - 4a^2\cos^2\theta\right] d\theta$$

$$= \frac{1}{2}\int_0^{\pi} 12a^2\cos^2\theta\,d\theta$$

$$= \frac{1}{2} \times 12a^2 \int_0^{\pi} \cos^2\theta\,d\theta$$

$\cos 2A = 1 - 2\sin^2 A$

$\cos 2A = 2\cos^2 A - 1$

$$\cos^2 A = \left[\frac{1+\cos 2A}{2}\right]$$

$$= 6a^2 \int_0^{\pi} \frac{1}{2}\left[1+\cos 2\theta\right] d\theta$$

$$= \frac{6a^2}{2}\left[\theta + \frac{\sin 2\theta}{2}\right]_0^{\pi}$$

$\int 1\, dx = x$

$$= 3a^2\left[\pi + \frac{\sin 2\pi}{2} - 0 - \frac{\sin 0}{2}\right]$$

$\int 1\, d\theta = \theta$

$$= 3a^2\pi \left[\sin 2\pi = \sin 0 = 0\right]$$

Session-21

Evaluation of triple Integrals

$$I = \int_e^f \int_c^d \int_a^b f(x,y;z)\, dx\, dy\, dz$$

(or) $$\int_0^c \int_0^b \int_0^a f(x,y,z)\, dz\, dy\, dx$$

(or) $$\int_0^2 \int_0^1 \int_{-1}^1 f(x,y,z)\, dy\, dx\, dz$$

1) Evaluate $\int_{x=0}^{1} \int_{y=0}^{2} \int_{z=0}^{3} xy^2 z\, dx\, dy\, dz$

Sol:

$\int x^n dx = \frac{x^{n+1}}{n+1}$

$$I = \int_{x=0}^{x=1} \int_{y=0}^{y=2} \int_{z=0}^{z=3} xy^2 z\, dz\, dy\, dx$$

$$= \int_0^1 \int_0^2 xy^2 \left[\frac{z^2}{2}\right]_0^3 dy\, dx$$

$$= \int_0^1 \int_0^2 xy^2 \, \frac{1}{2}\left[3^2 - 0^2\right] dy \, dx$$

$$= \frac{9}{2} \int_0^1 \int_0^2 xy^2 \, dy \, dx$$

$$= \frac{9}{2} \int_0^1 x \left[\frac{y^3}{3}\right]_0^2 dx$$

$$= \frac{9}{2} \int_0^1 x \left[\frac{8}{3} - 0\right] dx$$

$$= \frac{9}{2} \times \frac{8}{3} \int_0^1 x \, dx$$

$$= 12 \int_0^1 x \, dx$$

$$= 12 \left[\frac{x^2}{2}\right]_0^1$$

$$= 12 \left[\frac{1}{12} - 0\right] = \underline{6}$$

$$\int_0^1 \int_0^2 \int_0^3 xy^2 z \, dz \, dy \, dx$$

$$= \left[\int_0^1 x \, dx\right] \times \left[\int_0^2 y^2 dy\right] \times \left[\int_0^3 z \, dz\right]$$

$$= \left[\frac{x^2}{2}\right]_0^1 \times \left[\frac{y^3}{3}\right]_0^2 \times \left[\frac{z^2}{2}\right]_0^3$$

$$= \left[\frac{x^2}{2} - 0\right]_0^1 \times \left[\frac{8}{3} - 0\right] \times \left[\frac{9}{2} - 0\right]$$

$$= \frac{1}{2} \times \frac{8}{3} \times \frac{9}{2}$$

$$= \underline{6}$$

2) Evaluate $\iiint_B 8xyz \, dV$. where

B: $2 \le x \le 3$; $1 \le y \le 2$; and $0 \le z \le 1$

Sol:

$dV = dx \, dy \, dz$

$dz \, dy \, dx$

$$\int_{x=2}^{x=3} \int_{y=1}^{y=2} \int_0^1 8xyz \, dz \, dy \, dx$$

$$= 8\left[\int_{x=2}^{3} x \, dx\right] \times \left[\int_1^2 y \, dy\right] \times \left[\int_0^1 z \, dz\right]$$

$$= 8\left[\frac{x^2}{2}\right]_2^3 \times \left[\frac{y^2}{2}\right]_1^2 \times \left[\frac{z^2}{2}\right]_0^1$$

$$= 8\left[\frac{1}{2}[9-4]\right] \times \left[\frac{1}{2}(4-1)\right] \times \left[\frac{1}{2} - 0\right]$$

$$= 8\left[\frac{5}{2}\right] \times \left[\frac{3}{2}\right] \times \left[\frac{1}{2}\right]$$

$$= 15$$

3) Evaluate $\int_{x=0}^{x=a} \int_{y=0}^{y=x} \int_{z=0}^{z=x+y} e^{x+y+z} \, dz \, dy \, dx$

Sol:

$$\int_0^a \int_0^x \int_0^{x+y} e^{(x+y+z)} \, dz \, dy \, dx$$

$$= \int_0^a \int_0^x \left[e^{(x+y)+z}\right]_0^{x+y} dy \, dx$$

$$= \int_0^a \int_0^x \left[e^{(x+y)+(x+y)} - e^{(x+y)+0}\right] dy \, dx$$

$\int e^x \, dx = e^x$

$\int e^{ax} \, dx = \frac{e^{ax}}{a}$

$\int e^{x+1} \, dx = e^{x+1}$

$\int e^{ax} \, dx = \frac{e^{ax}}{a}$

$$= \int_0^a \int_0^x \left[e^{(2x+2y)} - e^{(x+y)} \right] dy\, dx$$

$$= \int_0^a \left[\frac{e^{(2x+2y)}}{2} - e^{(x+y)} \right]_0^x dx$$

$$= \int_0^a \frac{1}{2} \left[\frac{1}{2} e^{2x+2x} - e^{x+x} - \frac{1}{2} e^{2x+2\times 0} + e^{(x+0)} \right] dx$$

$$= \int_0^a \left[\frac{1}{2} e^{4x} - e^{2x} - \frac{1}{2} e^{2x} + e^{x} \right] dx$$

$$= \int_0^a \left[\frac{1}{2} e^{4x} - \frac{3}{2} e^{2x} + e^{x} \right] dx$$

$$= \left[\frac{1}{2} \frac{e^{4x}}{4} - \frac{3}{2} \frac{e^{2x}}{2} + e^{x} \right]_0^a$$

$$= \left[\frac{1}{8} e^{4a} - \frac{3}{4} e^{2a} + e^{a} - \frac{1}{8} e^{0} + \frac{3}{4} e^{0} - e^{0} \right]$$

$$= \left[\frac{1}{8} e^{4a} - \frac{3}{4} e^{2a} + e^{a} - \frac{1}{8} + \frac{3}{4} - 1 \right]$$

4) Evaluate $\int_1^e \int_1^{\log y} \int_1^{e^x} 1 \cdot dz\, dx\, dy$.

Sol:

$$I = \int_1^e \int_1^{\log y} \int_1^{e^x} 1 \cdot dz\, dx\, dy.$$

$$= \int_1^e \int_1^{\log y} [z]_1^{e^x} dx\, dy$$

$$= \int_1^e \int_1^{\log y} [e^x - 1]\, dx\, dy$$

$$= \int_1^e [e^x - x]_1^{\log y} dy.$$

$$= \int_1^e \left[e^{\log y} - \log y - e + 1 \right] dy$$

$$= \int_1^e \left[y - \log y - e + 1 \right] dy$$

$$= \int_1^e \left[(1-e) - y - \log y \right] dy \qquad \int c\,dx = cx$$

$$= \int_1^e (1-e)\,dy + \int_1^e y\,dy - \int_1^e \log y\,dy$$

$$= \left[[1-e]y \right]_1^e + \left[\frac{y^2}{2} \right]_1^e - \int_1^e \log y\,dy$$

$$I_3 = \int_1^e \log y\,dy$$

$$= \int_1^e \frac{1}{I} \cdot \log \frac{y}{I}\,dy$$

$$= (\log y)(y) - \int \frac{1}{y}\, y\,dy$$

$$= y \log y - \int 1\,dy$$

$$= \left[y \log - y \right]_1^e$$

$$= \left[e \log e - e - 1 \log 1 + 1 \right]$$

$$= 1$$

$$I_1 = \left[(1-e)y \right]_1^e$$

$$= \left[(1-e)e - (1-e)1 \right]$$

$$= (1-e)(e-1)$$

$$= -(e-1)^2$$

$I_2 = \frac{1}{2}\left[e^2 - 1\right]$

$I_3 = 1$

$I_1 + I_2 + I_3 = -(e-1)^2 + \frac{1}{2}(e^2-1) + 1$

Session - 22

Applications of triple integrals

volume of a solid over region V

$$V = \iiint_V dV.$$

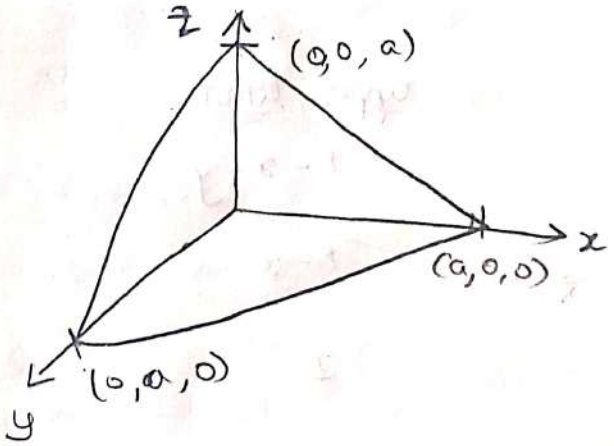
$dV = dz\ dy\ dx$ (or)

$dx\ dy\ dz$ (or)

$dy\ dx\ dz$.

$x + y + z = a$

Intercept form $\frac{x}{a} + \frac{y}{a} + \frac{z}{a} = 1$



Volume we are considering in first octant only.

So all variable's lower limit will be zero.

Upper limits:-

Equation of plane

$$x + y + z = a$$

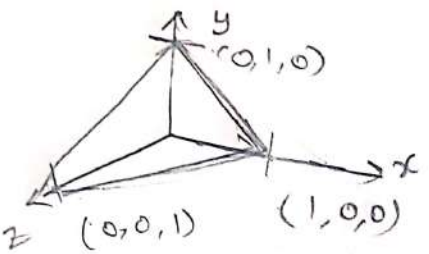
Upper limit of $z \Rightarrow z = a - x - y$

upper limit of $y \Rightarrow y = a - x$ { keeping $z = 0$ }

upper limit of $x \Rightarrow x = a$ { keeping $y = 0$ }

2) The planes are $x + y + z = 1$

Sol:-



The solid is bounded by coordinate planes

that is $x = 0$, $y = 0$ & $z = 0$

$$\text{Volume} = \iiint dv$$

$$= \iiint dz\, dy\, dx$$

Lower limit	Upper limit
$z : 0$	$1 - x - y$
$y : 0$	$1 - x$
$x : 0$	1

$$v = \int_0^1 \int_0^{1-x} \int_0^{1-x-y} dz\, dy\, dz$$

$$= \int_0^1 \int_0^{1-x} [z]_0^{1-x-y}\, dy\, dx$$

$$= \int_0^1 \int_0^{(1-x)} [(1-x) - y]\, dy\, dx$$

$$= \int_0^1 \left[(1-x)y - \frac{y^2}{2}\right]_0^{(1-x)} dx$$

$$= \int_0^1 \left[(1-x)(1-x) - \frac{1}{2}(1-x)^2 - 1(1-x)0 - \frac{0}{2}\right] dx$$

$$= \int_0^1 \left[(1-x)^2 - \frac{1}{2}(1-x)^2\right] dx$$

$$= \int_0^1 \frac{1}{2}\left[(1-x)^2\right] dx$$

$$= \frac{1}{2}\left[\frac{(1-x)^3}{(-1)3}\right]_0^1$$

$$\left\{ \int (ax+b)^n dx = \frac{1}{a}\frac{(ax+b)^{n+1}}{(n+1)} \right\}$$

$$= \frac{1}{2}\left(\left(\frac{-1}{3}\right)\cdot\left\{0^3 - 1^3\right\}\right)$$

$$= \frac{1}{6} \text{ cubic units}$$

13/8/26 Friday

Volume $= \iiint dv$

$dv = dz\, dy\, dx$

$z = z=0$ to $z = a-x-y$

$y = y=0$ to $y = a-x$

$x = x=0$ to $x = a$

$$V = \int_0^a \int_0^{a-x} \int_0^{a-x-y} dz\, dy\, dx$$

$$= \int_0^a \int_0^{a-x} [z]_0^{a-x-y}\, dy\, dx$$

$$= \int_0^a \int_0^{a-x} [(a-x)-y]\, dy\, dx$$

$$\boxed{\int (ax+b)^n dx = \frac{1}{a}\frac{(ax+b)^{n+1}}{(n+1)}}$$

$$= \int_0^a \left[(a-x)y - \frac{y^2}{2}\right]_0^{a-x} dx$$

$$= \int_0^a \left[(a-x)(a-x) - \frac{1}{2}(a-x)^2 - (a-x)0 - \frac{0^2}{2}\right] dx$$

$$= \int_0^a \left[(a-x)^2 - \frac{1}{2}(a-x)^2\right] dx$$

$$= \int_0^a \frac{1}{2}(a-x)^2\, dx$$

$$= \frac{1}{2}\left[\frac{(a-x)^3}{3(-1)}\right]_0^a$$

$$= -\frac{1}{6}\left[(a-a)^3 - (a-0)^3\right]$$

$$= -\frac{1}{6}\left[0^3 - a^3\right]$$

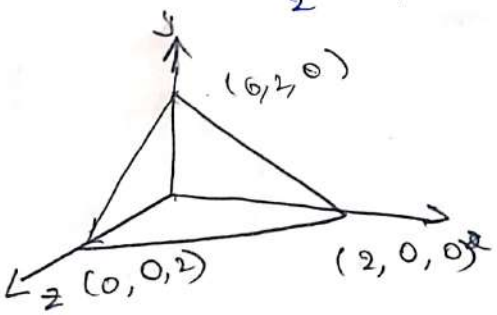
$$= \frac{1}{6}a^3 \text{ cubic units.}$$

Tutorial-9

5) tetrahedron bounded by the planes $x=0$ $y=0$, $z=0$. and $x+y+z=2$

Sol:-

$$\frac{x}{2} + \frac{y}{2} + \frac{z}{2} = 1$$



Limits of z

z: $z=0$ to $z=2-x-y$

y: $y=0$ to $y=2-x$

x: $x=0$ to $x=2$

$$V = \int_0^2 \int_0^{2-x} \int_0^{2-x-y} dz\, dy\, dx$$

$$= \int_0^2 \int_0^{2-x} [z]_0^{2-x-y}\, dy\, dx$$

$$= \int_0^2 \int_0^{2-x} [(2-x)-y]\, dy\, dx$$

$$= \int_0^2 \left[(2-x)y - \frac{y^2}{2}\right]_0^{2-x} dx$$

$$= \int_0^2 \left[(2-x)(2-x) - \frac{1}{2}(2-x)^2 - (2-x)0 - \frac{0^2}{2}\right] dx$$

$$= \int_0^2 \left[(2-x^2 - \frac{1}{2}(2-x)^2\right] dx$$

$$= \int_0^2 \left[\frac{-(2-x)^3}{3(-1)}\right]_0^2$$

$$= \frac{1}{-6}\left[(2-2)^3 - (2-0)^3\right]$$

$$= \frac{1}{6}\left[0^3 - 2^3\right] \qquad [0-8]$$

$$= \frac{1}{6}[8]$$

$$= \frac{8}{6} = \frac{4}{3}$$

4) planes & $x+y+z=8$.

Sol:- $dV = dz\, dy\, dx$

z: $z=0$ to $z=8-x-y$

y: $y=0$ to $y=8-x$

x: $x=0$ to $x=8$

$$V = \int_0^8 \int_0^{8-x} \int_0^{8-x-y} dz\, dy\, dx$$

$$= \int_0^8 \int_0^{8-x} [z]_0^{8-x-y} \, dy \, dx$$

$$= \int_0^8 \int_0^{8-x} [(8-x)-y] \, dy \, dx$$

$$= \int_0^8 \left[(8-x)y - \frac{y^2}{2} \right]_0^{8-x} dx$$

$$= \int_0^8 \left[(8-x)(8-x) - \frac{1}{2}(8-x)^2 - (8-x)0 - \frac{0^2}{2} \right] dx$$

$$= \int_0^8 \left[(8-x)^2 - \frac{1}{2}(8-x)^2 \right] dx$$

$$= \int_0^8 \frac{1}{2}(8-x)^2 dx$$

$$= \frac{1}{2}\left[\frac{(8-x)^3}{3(-1)} \right]_0^8$$

$$= -\frac{1}{6}\left[(8-8)^3 - (8-0)^3 \right]$$

$$= -\frac{1}{6}\left[0^3 - 512 \right]$$

$$= \frac{512}{6}$$

$$= \frac{256}{3} = 85.3$$

(2) Evaluate $\int_1^2 \int_2^3 \int_4^5 xyz \, dz \, dy \, dx$

Sol:-

$$\left[\int_1^2 x \, dx \right] \times \left[\int_2^3 y \, dy \right] \times \left[\int_4^5 z \, dz \right]$$

$$= \left[\frac{x^2}{2} \right]_1^2 \times \left[\frac{y^2}{2} \right]_2^3 \times \left[\frac{z^2}{2} \right]_4^5$$

$$= \frac{1}{2}(4-1) \times \frac{1}{2}(9-4) \times \frac{1}{2}(25-16)$$

$= \frac{1}{8} \times 3 \times 5 \times 9$

$= \frac{135}{8}$

③ Evaluate $\int_{0}^{x=1} \int_{y=0}^{y=x} \int_{z=0}^{z=x+y} xyz \, dz\, dy\, dx$

Sol:

$I = \int_0^1 \int_0^x \int_0^{x+y} xyz \, dz\, dy\, dx$

$= \int_0^1 \int_0^x xy \left[\frac{z^2}{2} \right]_0^{(x+y)} dy\, dx$

$= \int_0^1 \int_0^x xy \left[\frac{(x+y)^2}{2} - \frac{0}{2} \right] dy\, dx$

$= \frac{1}{2} \int_0^1 \int_0^x x\, y \left[x^2 + y^2 + 2xy \right] dy\, dx$

$= \frac{1}{2} \int_0^1 \int_0^x x \left[x^2 y + y^3 + 2xy^2 \right] dy\, dx$

$= \frac{1}{2} \int_0^1 x \left[x^2 \frac{y^2}{y} + \frac{y^4}{4} + 2x \frac{y^3}{3} \right]_0^x dx$

$= \frac{1}{2} \int_0^1 x \left[\frac{x^2}{2}(x^2) + \frac{1}{4}(x^4) + \frac{2x}{3}(x^3) - 0 - 0 - 0 \right] dx$

$= \frac{1}{2} \int_0^1 \left[\frac{x^5}{2} + \frac{1}{4} x^5 + \frac{2}{3} x^5 \right] dx$

$= \frac{1}{2} \int_0^1 \left[\frac{1}{2} + \frac{1}{4} + \frac{2}{3} \right] x^5 \, dx$

$= \frac{1}{2} \left[\frac{1}{2} + \frac{1}{4} + \frac{2}{3} \right] \left[\frac{x^6}{6} \right]_0^1$

$= \frac{1}{12} \left[\frac{1}{2} + \frac{1}{4} + \frac{2}{3} \right]$

(1) $\int_0^1 \int_0^2 \int_0^3 (x+y+z)\,dx\,dy\,dz$

Sol:- $\int_0^1 \int_0^2 \left[\frac{x^2}{2} + (y+z)\,x\right]_0^3 dy\,dz$

$= \int_0^1 \int_0^2 \left[\frac{9}{2} + 3(y+z) - 0 - 0\right] dy\,dz$

$= \int_0^1 \int_0^2 \left[\frac{9}{2} + 3y + 3z\right] dy\,dz$

$= \int_0^1 \left[\frac{9}{2}y + 3\cdot\frac{y}{2} + 3zy\right]_0^2 dz$

$= \int_0^1 \left[\frac{9}{2}(2) + \frac{3}{2}(2)^2 + 3z(2) - 0 - 0 - 0\right] dz$

$= \int_0^1 \left[9 + 6 + 6z\right] dz$

$= \int_0^1 \left[15 + 6z\right] dz$

$= \left[15z + 6\left(\frac{z^2}{2}\right)\right]_0^1$

$= \left[15(1) + 6\left(\frac{1}{2}\right) - 0 - 0\right]$

$= \underline{18}$

16/3/26
Monday

CO-4

Vector calculus

Gradient of a scalar point funⁿ

Operator del $(\nabla) = \left(\hat{i}\frac{\partial}{\partial x} + \hat{j}\frac{\partial}{\partial y} + \hat{k}\frac{\partial}{\partial z}\right)$

gradient of a scalar point fun $f(x,y,z)$

$$\Delta f = \left(i\frac{\partial}{\partial x} + j\frac{\partial}{\partial y} + k\frac{\partial}{\partial z}\right) f(x,y,z)$$

$$\Delta f \text{ (or) } \text{grad } f = \left(i\frac{\partial f}{\partial x} + j\frac{\partial f}{\partial y} + k\frac{\partial f}{\partial z}\right)$$

(or)

gradient of f

Note:- g. and f (or) ∇f, [where $f(x,y,z)$ be any surface] indicates the normal vector of surface $f(x,y,z)$

Angle between two surface:-

Let $f(x,y,z)$ and $g(x,y,z)$ be two surface.

~~Lace~~ let the angle between them be 'θ'

$$\cos\theta = \frac{(\nabla f)\cdot(\nabla g)}{|\nabla f|\,|\nabla g|}$$

Direction Derivative :-

The Dirⁿ Derivative for any surface / scalar pt fun $f(x,y,z)$ in dirⁿ of $\vec{a}$, can be given by,

$$\text{Direction Derivative} = \frac{(\nabla f)\cdot\vec{a}}{|\vec{a}|}$$

(or) $\nabla f \cdot \hat{a}$

Session- 23:-

① determine grad f where $f = x^3 - y^3 + 3xyz$ at the Point (1, 2, 3).

Sol:

$$\text{grad } f = i\frac{\partial f}{\partial x} + j\frac{\partial f}{\partial y} + k\frac{\partial f}{\partial z}$$

$$f(x, y, z) = x^3 - y^3 + 3xyz$$

$$\frac{\partial f}{\partial x} = 3x^2 + 3yz \; ; \; \frac{\partial f}{\partial y} = -3y^2 + 3xz$$

$$\frac{\partial f}{\partial z} = 3xy$$

$$\nabla f = [3x^2 + 3yz]\hat{i} + [-3y^2 + 3xz]\hat{j} + (3xy)\hat{k}$$

$$(\nabla f)_{(1,2,3)} = [3(1)^2 + 3(2)(3)]\hat{i} + (-3(2)^2 + 3(1)(3)]\hat{j} + [3(1)(2)]\hat{k}$$

$$= 21\hat{i} - 3\hat{j} + 6\hat{k}$$

② The temperature point at (x, y, z) by $T(x, y, z) = x^2 + y^2 - z$ a mosquito located at (1, -1, 2).

Sol:

as the height ↑ temp will ↓ so, the mosquito should fly normal dirn to that surface which is grad of surface

$$T(x, y, z) = x^2 + y^2 - z$$

normal vector = ∇T (or) grad T

$$\nabla T = \hat{i}\frac{\partial T}{\partial x} + \hat{j}\frac{\partial T}{\partial y} + \hat{k}\frac{\partial T}{\partial z}$$

$$\nabla T = (2x)\hat{i} + (2y)\hat{j} - 1$$

$(\nabla i)(1,-1,2) = 2\hat{i} - 2\hat{j} - k$

③ $c(x,y,z) = x^2y - 2xz - 4$ & unit vector

Sol: At the point 1, -1, 0.

$c(x,y,z) = x^2y - 2xz - 4$

unit normal vector $= \dfrac{\nabla c}{|\nabla c|}$

$$\nabla c = \hat{i}\frac{\partial c}{\partial x} + \hat{j}\frac{\partial c}{\partial y} + \hat{k}\frac{\partial c}{\partial z}$$

$$\frac{\partial c}{\partial x} = (2xy - 2z); \quad \frac{\partial c}{\partial y} = x^2; \quad \frac{\partial c}{\partial z} = 2x$$

$$\nabla c = (2xy - 2z)\hat{i} + (x^2)\hat{j} + (-2x)\hat{k}$$

$(\nabla c)_{(1,-1,0)} = (-2-0)\hat{i} + (1^2)\hat{j} + (-2(1))\hat{k}$

$= -2\hat{i} + \hat{j} - 2\hat{k}$

$|\nabla c| = \sqrt{2^2 + 1^2 + 2^2} = \sqrt{9} = 3$

unit normal vector $= \dfrac{\nabla c}{|\nabla c|} = \dfrac{1}{3}(2\hat{i} + \hat{j} - 2\hat{k})$

⑤ compute the direction derivative of $f = [3x + 2y - 2z]$ in the direction of vector $2\hat{i} + \hat{j} + 3\hat{k}$ at the point (1,1,1)

Sol: $f = [3x + 2y - 2z]$

$\vec{a} = 2\hat{i} + \hat{j} + 3\hat{k}$

Directional deriv $= \dfrac{\nabla f \cdot \vec{a}}{|\vec{a}|}$

$$\nabla f = \hat{i}\frac{\partial f}{\partial x} + \hat{j}\frac{\partial f}{\partial y} + k\frac{\partial f}{\partial z}$$

$(\nabla f) = 3\hat{i} + 2\hat{j} - 2\hat{k}$

$(\nabla f)_{(1,1,1)} = 3\hat{i} + 2\hat{j} - 2\hat{k}$

$DD = \dfrac{(\nabla f)\cdot\vec{a}}{|\vec{a}|}$

$$= \frac{(3\hat{i} + 2\hat{j} - 2\hat{k}) \cdot (2\hat{i} + \hat{j} + 3\hat{k})}{(\sqrt{14})}$$

$$= \frac{6+2-6}{\sqrt{14}} = 2/\sqrt{14}$$

⑥ Find the dir^n derivative of $\varphi = xy - yz - zx$ at A in the direction of $\vec{AB}$; where $A(1,2,0)$ & $B(1,0,3)$.

Sol^n: $\vec{AB} = (P.V.\ \vec{B}) - (P.V.\ \vec{A})$

$$= (\hat{i} + 0\hat{j} + 3\hat{k}) - (\hat{i} + 2\hat{j} + 0\cdot\hat{k})$$

$$\vec{AB} = 0\cdot\hat{i} - 2\hat{j} + 3\hat{k}$$

$$\varphi = xy - yz - zx$$

$$DD = \nabla\varphi \cdot \frac{\vec{AB}}{|\vec{AB}|}$$

$$\nabla\varphi = \hat{i}\frac{\partial\varphi}{\partial x} + \hat{j}\frac{\partial\varphi}{\partial y} + k\frac{\partial\varphi}{\partial z}$$

$$\frac{\partial\varphi}{\partial x} = y - z; \quad \frac{\partial\varphi}{\partial y} = x - z; \quad \frac{\partial\varphi}{\partial z} = -y$$

$$\nabla\varphi = (y-z)\hat{i} + (x-z)\hat{j} + (-y-x)\hat{k}$$

$$(\nabla\varphi)_{(1,2,0)} = 2\hat{i} + \hat{j} - 3\hat{k}$$

$$DD = \frac{(2\hat{i} + \hat{j} - 3\hat{k})(0\hat{i} - 2\hat{j} + 3\hat{k})}{\sqrt{13}} = \frac{-9}{\sqrt{13}} = \frac{-11}{\sqrt{13}}$$

⑦ 2 geological layers in earth's crust $x^2 + y^2 + z^2 = 14$ at the pt $P(1,2,3)$ and $x^2 + 2y^2 + z^2 = 12$ at the pt $Q(2,2,2)$ Determine the angle between two layers.

Sol:- consider $f = x^2 + y^2 + z^2 = 14$;

$g = x^2 + 2y^2 + z^2 = 12$;

$$\cos\theta = \frac{(\nabla f)\cdot(\nabla g)}{|\nabla f|\cdot|\nabla g|}$$

$$\nabla f = i\frac{\partial f}{\partial x} + j\frac{\partial f}{\partial y} + k\frac{\partial f}{\partial z}$$

$$= (2x)\hat{i} + (2y)\hat{j} + (2z)\hat{k}$$

$$(\nabla f)_{(1,2,3)} = 2\hat{i} + 4\hat{j} + 6\hat{k}$$

$$\nabla g = \hat{i}\frac{\partial g}{\partial x} + \hat{j}\frac{\partial g}{\partial y} + \hat{k}\frac{\partial g}{\partial z}$$

$$= (2x)\hat{i} + (4y)\hat{j} + (2z)\hat{k}$$

$$(\nabla g)_{(2,2,2)} = 4\hat{i} + 8\hat{j} + 4\hat{k}$$

$$\cos\theta = \frac{(\nabla f)\cdot(\nabla g)}{|\nabla f||\nabla g|}$$

$$= \frac{(2\hat{i} + 4\hat{j} + 6\hat{k})\cdot(4\hat{i} + 8\hat{j} + 4\hat{k})}{\sqrt{4+16+36}\cdot\sqrt{16+64+16}}$$

$$= \frac{8 + 32 + 24}{\sqrt{56}\ \sqrt{96}}$$

$$= \frac{64}{\sqrt{56}\cdot\sqrt{96}} \text{ (or) } \theta = \cos^{-1}\left(\frac{64}{\sqrt{56}\cdot\sqrt{96}}\right)$$

Session-24:-

Divergence & curl of a vector point fun^n & (or) vector fn

Divergence of a vector pt fun:

Let $\vec{F} = f_1\hat{i} + f_2\hat{j} + f_3\hat{k}$ be the vector field Then the divergence of $\vec{F}$ is given by ($\nabla\cdot\vec{F}$ (or) div $\vec{F}$)

$$\nabla \vec{f} = \left(\hat{i}\frac{\partial}{\partial x} + \hat{j}\frac{\partial}{\partial y} + \hat{k}\frac{\partial}{\partial z}\right)\cdot\left(f_1\hat{i} + f_2\hat{j} + f_3\hat{k}\right)$$

$$\nabla\cdot\vec{f} = \frac{\partial f_1}{\partial x} + \frac{\partial f_2}{\partial y} + \frac{\partial f_3}{\partial z}$$

23/3/26
Monday

Divergence of a vector point fun:-

$\nabla\cdot\vec{v}$, $\vec{v} = v_1\hat{i} + v_2\hat{j} + v_3\hat{k}$

$$\nabla\cdot\vec{v} = \frac{\partial v_1}{\partial x} + \frac{\partial v_2}{\partial y} + \frac{\partial v_3}{\partial z}$$

23/3/26
Monday

Divergence of a vector point fun:-

$\nabla \cdot \vec{v}$, $\vec{v} = v_1 \bar{i} + v_2 \bar{j} + v_3 \bar{k}$

$$\nabla \cdot \vec{v} = \frac{\partial v_1}{\partial x} + \frac{\partial v_2}{\partial y} + \frac{\partial v_3}{\partial z}$$

Note:- If div $\vec{v}$ (or) $\nabla \cdot \vec{v} = 0$, then $\vec{v}$ is a solenoidal vector.

Curl of a vector point fun / vector field:-

It is the cross product of operator well and vector v.

- It is denoted as $\nabla \times \vec{v}$ (or) curl $\vec{v}$ and defined as

$$\vec{\nabla} \times \vec{v} = \begin{vmatrix} i & j & k \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ v_1 & v_2 & v_3 \end{vmatrix}$$

$$\bar{i}\left[\frac{\partial v_3}{\partial y} - \frac{\partial v_2}{\partial z}\right] - \bar{j}\left[\frac{\partial v_3}{\partial x} - \frac{\partial v_1}{\partial z}\right] + \bar{k}\left[\frac{\partial v_2}{\partial x} - \frac{\partial v_1}{\partial y}\right]$$

Note:- If curl $\vec{v} = \vec{0}$, then $\vec{v}$ is irrotational

Session-24

① $\vec{f} = 2x^2\bar{i} + 4y^2\bar{j} + 6z^2\bar{k}$ at the point (1,1,2).

Sol:- $\vec{v} = v_1 \hat{i} + v_2 \hat{j} + v_3 \bar{k}$

$$\nabla \cdot \vec{v} = \frac{\partial v_1}{\partial x} + \frac{\partial v_2}{\partial y} + \frac{\partial v_3}{\partial z}$$

$v_1 = 2x^2$; $v_2 4y^2$, $v = z^2$

$\nabla \cdot v = 4x + 8y + 12z$

$(\nabla \cdot v)(1,1,2) = 4(1) + 8(1) + 12(2)$

$= 4 + 8 + 24$

$= \underline{\underline{36}}$.

② Determine P; if $\vec{F} = (x + 3y)\hat{i} + (y - 2z)\hat{j} + (x + pz)\hat{k}$ is solenoidal vector.

Sol: Given $\vec{F}$ is solenoidal so div $\vec{F} = 0$

$\vec{F} = (x+3y)\hat{i} + (y - 2z)\hat{j} + (x + pz)\hat{k}$

$\Rightarrow$ div $\vec{F} = 0$

$$\frac{\partial}{\partial x}(x + 3y) + \frac{\partial}{\partial y}(y - 2z) + \frac{\partial}{\partial z}(x + pz) = 0$$

$\Rightarrow 1 + 1 + p = 0$

$\Rightarrow \underline{\underline{p = -2}}$

③ A wind velocity field is given by $\vec{v} = (y^2, -x, z^3)$ find the curl of the field.

Sol: Given

$$\vec{v} = \underbrace{y^2}_{V_1}\hat{i} + \underbrace{(-x)}_{V_2}\hat{j} + \underbrace{z^3}_{V_3}\hat{k}$$

$$\text{Curl } \vec{v} = \begin{vmatrix} \hat{i} & \hat{j} & k \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ V_1 & V_2 & V_3 \end{vmatrix}$$

$$\text{Curl } \vec{v} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ y^2 & -x & z^3 \end{vmatrix}$$

$$= \hat{i}\left[\frac{\partial}{\partial y}(z^3) - \frac{\partial}{\partial z}(-x)\right] - \hat{j}\left[\frac{\partial}{\partial x}(z^3) - \frac{\partial}{\partial z}(y^2)\right] +$$

$$\hat{k}\left[\frac{\partial}{\partial x}(-x) - \frac{\partial}{\partial y}(y^2)\right]$$

$$= \hat{i}(0-0) - \hat{j}(0-0) + \hat{k}[-1-2y]$$

$$= (-2y-1)\hat{k}$$

(4) $\vec{v} = (y, -x, 0)$ and determine whether the flow is rotational.

Sol: $\vec{v} = \underset{v_1}{y}\hat{i} + \underset{v_2}{(-x)}\hat{j} + \underset{v_3}{0}\hat{k}$

$$\text{curl } \vec{v} = \begin{vmatrix} i & j & k \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ v_1 & v_2 & v_3 \end{vmatrix}$$

$$\text{curl } \vec{v} = \begin{vmatrix} i & j & k \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ y & -x & 0 \end{vmatrix}$$

$$= i\left[\frac{\partial}{\partial y}(0) - \frac{\partial}{\partial z}(-x)\right] - j\left[\frac{\partial}{\partial x}(0) - \frac{\partial}{\partial z}(y)\right] + k\left[\frac{\partial}{\partial x}(-x) - \frac{\partial}{\partial y}(y)\right]$$

$$= i[0-0] - j[0-0] + k[-1-1] \Rightarrow -2k$$

$\vec{v}$ is not irrotational $\because$ curl $\vec{v} \neq \vec{0}$

$$= i\left[\frac{\partial}{\partial y}(z^3) - \frac{\partial}{\partial z}(-x)\right] - j\left[\frac{\partial}{\partial x}(z^3) - \frac{\partial}{\partial z}(y^2)\right] +$$

$$k\left[\frac{\partial}{\partial x}(-x) - \frac{\partial}{\partial y}(y^2)\right]$$

$$= i(0-0) - j(0-0) + k[-1-2y]$$

$$= (-2y-1)\hat{k}$$

(4) $\vec{v} = (y, -x, 0)$. and determine whether the flow is rotational.

Sol: $\vec{v} = \underset{v_1}{y\hat{i}} + \underset{v_2}{(-x)\hat{j}} + \underset{v_3}{0\hat{k}}$

$$\text{curl } \vec{v} = \begin{vmatrix} i & j & k \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ v_1 & v_2 & v_3 \end{vmatrix}$$

$$\text{curl } \vec{v} = \begin{vmatrix} i & j & k \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ y & -x & 0 \end{vmatrix}$$

$$= i\left[\frac{\partial}{\partial y}(0) - \frac{\partial}{\partial z}(-x)\right] - j\left[\frac{\partial}{\partial x}(0) - \frac{\partial}{\partial z}(y)\right] + k\left[\frac{\partial}{\partial x}(-x) - \frac{\partial}{\partial y}(y)\right]$$

$$= i[0-0] - j[0-0] + k[-1-1] \Rightarrow -2k$$

$\vec{v}$ is not irrotational $\because$ curl $\vec{v} \neq \vec{0}$

⑥ determine a, b and c if $\vec{f} = (2x + 3y + az)\bar{i} + (bx + 2y + 3z)\bar{j} + (2x + 6y + 3z)\bar{k}$ is rotational

Sol:- given $\vec{f}$ is irrotational so curl $\vec{f} = \vec{0}$

$$\text{curl } \vec{f} = \begin{vmatrix} i & j & k \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ 2x+3y & bx+2y & 2x+ \\ az & +3z & 6y+3z \end{vmatrix}$$

$$\Rightarrow i\left[\frac{\partial}{\partial y}(2x + 6y + 3z) - \frac{\partial}{\partial z}[bx + 2y + 3z)\right] - \bar{j}$$

$$\left[\frac{\partial}{\partial x}(2x + (y + 3z)) - \frac{\partial}{\partial z}(2x + 3y + az)\right] +$$

$$\vec{k}\left[\frac{\partial}{\partial x}(bx + 2y + 3z) - \frac{\partial}{\partial y}(2x + 3y + az)\right.$$

$$\Rightarrow i[c-3] - \bar{j}[2-a] + \bar{k}[b-3]$$

$$\vec{0} \times \vec{F} = \vec{0} \Rightarrow \nabla \times \vec{F} = 0\bar{i} + 0\bar{j} + 0\bar{k}$$

$c - 3 = 0$ | $-2 + a = 0$ | $b - 3 = 0$

$\Rightarrow \boxed{c = 3}$ | $\boxed{a = 2}$ | $\boxed{b = 3}$

Tutorial 10 :-

① $f = 4x^2 + 3y^2 + 2z^2$ at the point $(1,1,1)$

$= 8x + 6y + 4z$

$(1,1,1) = 8(1) + 6(1) + 4(1)$

$= 18$

(2) $T(x,y,z) = 2x^2 - 3y + 4z$ at point $(1,1,-1)$

$= 4x - 3y + 4z$

$(1,1,-1) = 4(1) - 3(1) + 4(-1)$

$= 4 - 3 + 4$

$= 1 + 4 = +5$

(3) $V(x,y,z) = x^2 - yz$ at a point $(1,-1,2)$

(4)!-

28/3/26
Saturday

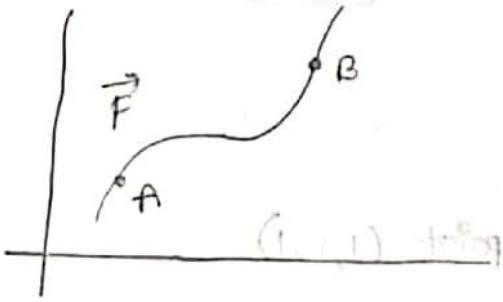
Session-25

The Integral, work done by force:

Work Computation:-

Let a particle moves along a path from point A to point B, and a force $\vec{F}$, acted on particle, then the work done by particle from moving A to B is

$$\omega = \int_C \vec{F} \cdot d\vec{r}$$



$\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$

$d\vec{r} = dx\hat{i} + dy\hat{j} + dz\hat{k}$

① compute the work done by the force $\vec{F} = 2xy\hat{i} - 3z\hat{j} + 5xk$ when it moves a particle along the curve $x = t^2+1$, $y = 2t^2$, $z = t^3$ from $t=1$ to $t=2$

Sol: $w = \int_C \vec{F} \cdot d\vec{r}$

$\vec{F} = 2xy\hat{i} - 3z\hat{j} + 5xk$

$d\vec{r} = dx\hat{i} + dy\hat{j} + dz\hat{k}$

$\vec{F} \cdot d\vec{r} = (2xy)dx - (3z)\,dy + (5x)dz$

given $x = t^2+1$

$\frac{dx}{dt} = 2t$

$dx = 2t\,dt$

$y = 2t^2$

$\frac{dy}{dt} = 4t$

$dy = 4t\,dt$

$z = t^3$

$\frac{dz}{dt} = 3t^2$

$dz = 3t^2 dt$

$\vec{F}d\vec{r} = [2(t^2+1)(2t^2).(2t\,dt)] + [-3(t^3)(4t\,dt)] + [5(t^2+1).3t^2dt]$

$= [8t^3(t^2+1) - 12t^4 + 15t^2(t^2+1)]\,dt$

$= [8t^5 + 8t^3 - 12t^4 + 15t^4 + 15t^2]\,dt$

$= [8t^5 + 3t^4 + 8t^3 + 15t^2]\,dt$

$= \frac{4}{3}[2^6-1] + \frac{3}{5}(2^5-1) + 2(2^4-1) + 5(2^3-1)$

work

$w = \int_{t=1}^{t=2} \vec{F} \cdot d\vec{r}$

$= \int^{t=2} [8t^5 + 3t^4 + 8t^3 + 15t^2]\,dt$

$\left[\frac{t^6}{6} + 3 \cdot \frac{t^5}{5} + 8\frac{t^4}{4} + 15\frac{t^3}{3}\right]_1^2$

$\left[t^6 + \frac{3}{5}t^5 + 2t^4 + 5t^3\right]_1^2$

② If $\vec{F} = 2xy\hat{i} - 3yz\hat{j} + 4xz\hat{k}$, Evaluate $\int_C \vec{F}\cdot d\vec{r}$ from $(0,0,0)$ to $(1,1,1)$, along the path $C: x = t;\ y = t^2;\ z = t^3$

Sol:-

$$\vec{F}\cdot dr = (2xy)\,dx - (3yz)\,dy + (4xz)\,dz$$

$$d\vec{r} = dx\hat{i} + dy\hat{j} + dz\hat{k}$$

Give,

$x = t$	$y = t^2$	$z = t^3$
$x = dt$	$y = 2t$	$z = 3t^2$
	$y = 2t\,dt$	$z = 3t^2\,dt$

$$= 2(t)(t^2)\cdot dt - 3(t^2)(t^3)2t\,dt + 4(t)(t^3)3t^2\,dt$$

$$= [2t^3 - 6t^6 + 12t^6]\,dt = (2t^3 + 6t^6)\,dt$$

$(0,0,0)$ to $(1,1,1)$

$x = t$

when $x = 0 \Rightarrow t = 0$

$x = 1 \Rightarrow t = 1$

limits $\boxed{t = 0 \text{ to } t = 1}$

$$\int_C \vec{F}\cdot dr = \int_{t=0}^{t=1} (2t^3 + 6t^6)\,dt$$

$$= \left[2\cdot\frac{t^4}{4} + 6\frac{t^7}{7}\right]_0$$

$$= \frac{1}{2} + \frac{6}{7}$$

(3) A conductor moves along the straight line from (0,0,0) to (2,1,3) in a magnetic field described by $\vec{F} = 2x^2\hat{i} + (4xz - y)\hat{j} + 2z\hat{k}$. Calculate the work done by the magnetic force using a line integral $\int_C \vec{F}\cdot d\vec{r}$

Sol: (x_1, y_1, z_1) to (x_2, y_2, z_2)

$$\frac{x - x_1}{x_2 - x_1} = \frac{y - y_1}{y_2 - y_1} = \frac{z - z_1}{z_2 - z_1} (= t)$$

$$\frac{x - 0}{2 - 0} = \frac{y - 0}{1 - 0} = \frac{z - 0}{3 - 0} (= t)$$

$\frac{x}{2} = t \Rightarrow x \Rightarrow 2t \Rightarrow dx = 2dt$

$\frac{y}{1} = t \Rightarrow y = t \Rightarrow dy = dt$

$\frac{z}{3} = t \Rightarrow z = 3t \Rightarrow dz = 3dt$

to get 't' limits:

$x = 2t$

when $x = 0 \Rightarrow t = 0$

$x = 2, \Rightarrow 2 = 2t \Rightarrow t = 1$

$\vec{F}\cdot d\vec{r}$

$= 2x^2 dx + (4xz - y)\, dy + 2z\, dz$

$\vec{F}\cdot d\vec{r} = 2(2t)^2 2(dt) + (4(2t)(3t) - t)(dt) + 2(3t).\, 3dt$

$= [16t^2 + (24t^2 - t) + 18t]\, dt$

$= [40t^2 + 17t]\, dt$

$\int_C \vec{F}\cdot d\vec{r}$

$= \int_0^1 (40t^2 + 17t)\, dt$

$= \left[40\left(\frac{t^3}{3}\right) + 17\left(\frac{t^2}{2}\right)\right]_0^1$

$= \frac{40}{3} + \frac{17}{2}$

$=$

(4) A drone moves along a curve path C: $y = 2x^2$ in the xy-plane under the wind force $\vec{F} = 3xy\vec{i} - y^2\vec{j}$, from (0,0) to (1,2). Find the work done by the wind using a line integral $\int_C \vec{F} \cdot d\vec{r}$

Sol:- ~~$\vec{F} \cdot d\vec{r} = (3xy)dx - (y^2)dy$~~

curved path C: $y = 2x^2$

$\vec{F} \cdot d\vec{r} = (3xy)dx - y^2 dy$

diff w.r.to 'x' $\frac{dy}{dx} = 4x$ (or) $dy = 4x\,dx$

$\vec{F} \cdot d\vec{r} = 3x(2x^2)dx - (2x^2)^2(4x\,dx)$

$= [6x^3 - 16x^5]dx$

x limits : $x = 0$ to $x = 1$

$\int_C \vec{F} \cdot d\vec{r} = \int_0^1 (6x^3 - 16x^5)dx$

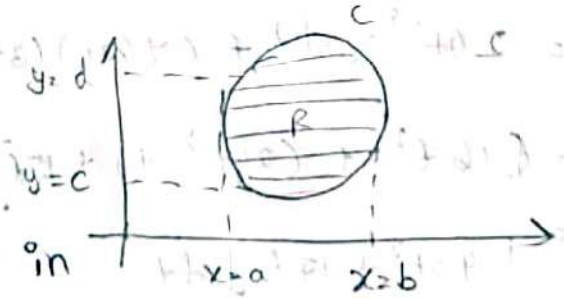
$= \left[6 \cdot \frac{x^4}{4} - 16 \cdot \frac{x^6}{6}\right]_0^1$

$= \left[\frac{3}{2} - \frac{8}{3} - 0 - 0\right]$

Session - 27:

Green's theorem and Its Applications:-

If $M(x,y)$ & $N(x,y)$ and their partial derivative $\frac{\partial M}{\partial y}$, $\frac{\partial N}{\partial x}$ are continuous in



Some region R of the xy plane, which is bounded by a closed curved C, then,

$$\oint_C [M.dx + N dy] = \iint_R \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}\right) dx\, dy$$

(1) Apply Green's theorem, to evaluate the integral $\int_C [(xy - y^2)\, dx - x^2 dy]$ where C is bounded by $y = x$ & $y = x^2$

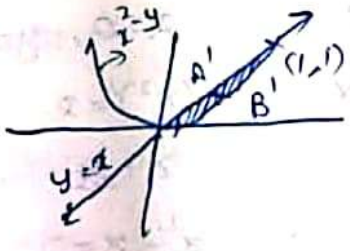
Sol: By Green's the

$$\int_C M.dx + N\, dy = \iint_R \left(\frac{dN}{dx} - \frac{\partial M}{\partial y}\right) dx\, dy$$

$M = xy - y^2$; $N = -x^2$

$\frac{\partial M}{\partial y} = x - 2y$; $\frac{\partial N}{\partial x} = -2x$

C is Bounded by $y = x$ & $y = x^2$



limits of x: $x = y$ to $x = \sqrt{y}$

y: $y = 0$ to $y = 1$

$$\int_C (xy - y^2)\, dx + (-x^2) dy = \int_0^1 \int_y^{\sqrt{y}} \{-2x - [x - 2y]\}\, dx\, dy$$

$$= \int_0^1 \int_y^{\sqrt{y}} (-3x + 2y)\, dx\, dy$$

$$= \int_0^1 \left[-3\frac{x^2}{2} + 2y\, x\right]_y^{\sqrt{y}} dy$$

$$= \int_0^1 \left[-\frac{3}{2}y + 2y(\sqrt{y}) + \frac{3}{2}y^2 - 2y(y) \right] dy$$

$$= \int_0^1 \left[-\frac{3}{2}y + 2y^{3/2} + \frac{3}{2}y^2 - 2y \right] dy$$

$$= \int_0^1 \left[2y^{3/2} - \frac{1}{2}y^2 - \frac{3}{2}y \right] dy$$

$$= 2\frac{y^{5/2}}{5/2} - \frac{1}{2}\left(\frac{y^3}{3}\right) - \frac{3}{2}\left(\frac{y^2}{2}\right)\Big]_0^1$$

$$= \left[\frac{4}{5} - \frac{1}{6} - \frac{3}{4} - 0 - 0 - 0 \right]$$

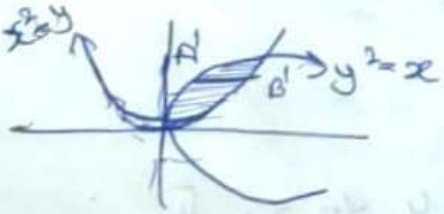
30/3/26
Monday

(2) Apply Green's theorem to evaluate the integral $\int_C [(2xy - x^2)dx + (x^2 + y^2)dy]$. where C is bounded by $y = x^2$ and $y^2 = x$.

Sol:- By Green's theorem,

$$\oint_C M dx + N dy = \iint_R \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) dx\, dy$$

$M = \frac{\partial M}{\partial y}$ $\qquad$ $N = \frac{\partial N}{\partial x}$



$0 \le x \le 1$

$x^2 \le y \le \sqrt{x}$

$M = 2x - x^2$ $\qquad$ $N = x + y^2$

$\frac{\partial M}{\partial y} = 2x$ $\qquad$ $\frac{\partial N}{\partial x} = 1$

$x^2 = y$

$(y^2)^2 = y$

$y^4 - y = 0$

$y(y^3 - 1) = 0$

~~$x=0, y=$~~

$(x^2)^2 = x$

$x^4 = x$

$x^4 - x = 0$

$x(x^3 - 1) = 0$

$x = 0, x = 1$

x are from 0 to 1

$x = 0, y = 0^2 = 0$

for $x = 1, y = 1^2 = 1$

$x = 0$ & $x = 1$

$y = \sqrt{x}$ (from $y^2 = x$)

$y = x^2$

y x^2 to $\sqrt{x}$

$$\int_0^1 \left[x - \cancel{2}\,\frac{x^2}{\cancel{2}}\right]_{y^2}^{\sqrt{y}} dy$$

$$= \int_0^1 \left[\sqrt{y} - y - y^2 + y^4\right] dy$$

$$= \left[\frac{y^{3/2}}{3/2} - \frac{y^2}{2} - \frac{y^3}{3} + \frac{y^5}{5}\right]_0^1 = \left[\frac{2}{3} - \frac{1}{2} - \frac{1}{3} + \frac{1}{5}\right] = \frac{1}{3} - \frac{1}{2} + \frac{1}{5}$$

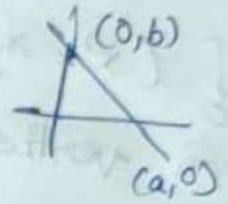
$$= \frac{10-15+6}{30} = \frac{1}{30}$$

③ Apply green's theorem, evaluate $\int_C [(3x - 8y^2)\,dx + (4y - 6xy)\,dy]$ where C is the boundary of the region bounded by $x=0, y=0$ and $x=0, y=0$ and $x+y=1$

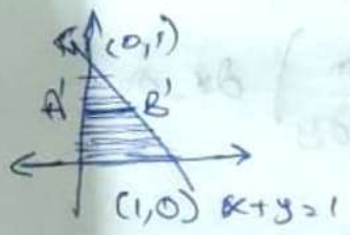
Sol:-

$M = 3x - 8y^2$; $N = 4y - 6xy$

$\frac{\partial M}{\partial y} = 16y$; $\frac{\partial N}{\partial x} = -6y$



$\frac{x}{a} + \frac{y}{b} = 1$



$$\oint M\,dx + N\,dy = \iint_R \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}\right) dx\,dy$$

x limits:- $x = 0$ to $x = 1-y$

y limits:- $y = 0$ to $y = 1$

$$\iint_R \frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}$$

$$= \int_0^1 \int_0^{1-y} [-6y - (-16y)]\,dx\,dy$$

$$= \int_0^1 \int_0^{1-y} 10y\,dx\,dy$$

$$= 10 \int_0^1 y\,(x)_0^{1-y}\,dy$$

$= 10 \int_0^1 y(1-y)\,dy$

$= 10 \int_0^1 (y - y^2)\,dy$

$= 10 \left[\frac{y^2}{2} - \frac{y^3}{3} \right]_0^1$

$= 10 \left[\frac{1}{2} - \frac{1}{3} - 0 - 0 \right]$

$= 10 \left[\frac{1}{6} \right]$

$= 5/3$

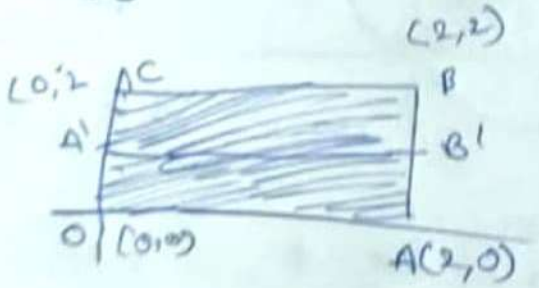
(4) $\int_C (x^2 - xy^3)\,dx + (y^2 - 2xy)\,dy$ where C is a square with vertices $(0,0), (2,0), (2,2)$ & $(0,2)$

Sol: By green's th:

$$\oint_C M\,dx + N\,dy = \iint_R \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) dx\,dy$$

$M = x^2 - xy^3$; $N = y^2 - 2xy$

$\frac{\partial M}{\partial y} = 3xy^2$, $\frac{\partial N}{\partial x} = -2y$



x limits: $x=0$ to $x=2$

y limits: $y=0$ to $y=2$

$$\iint_R \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) dx\,dy = \int_0^2 \int_0^2 \left[-2y - (3xy^2) \right] dx\,dy$$

$$= \int_0^2 \int_0^2 \left(-2y + 3xy^2 \right) dx\,dy$$

$$= \int_0^2 \left[-2xy + 3y^2\left(\frac{x^2}{2}\right)\right]_0^2 dy$$

$$= \int_0^2 \left[-2y(2) + 3y^2\left(\frac{4}{2}\right)\right] dy$$

$$= \int_0^2 (-4y + 6y^2)\, dy$$

$$= \left[-4\left(\frac{y^2}{2}\right) + 6\left(\frac{y^3}{3}\right)\right]_0^2$$

$$= \left[-2y^2 + 2y^3\right]_0^2$$

$$= \left[-2(4) + 2(8) - 0 - 0\right]$$

$$= [-8 + 16] = \underline{\underline{8}}$$